rexroth
A Bosch Company

Release Notes

G-Code Runtime App

G-Code Interpreter for ctrlX OS

Version 4.6.0

Edition 02

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DC-AE/PAD-SYS (MiSc/MePe)

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1 About this documentation

Editions of this documentation

Edition	Release date	Note
01	2025-11	G-Code Runtime App Version 4.4.0
02	2026-03	G-Code Runtime App Version 4.6.0

Further information

- See ➦ G-Code Runtime App, Application Manual

2 Important information

2.1 Product information

For the latest information on our products, go to ➦ https://www.boschrexroth.com.

For current security advisories, please refer to ➦ https://psirt.bosch.com/security-advisories/

3 Version 4.6.0

3.1 New functions

3.1.1 Feature 558941: Global variables with the character "$" and local variables without a preceding character can be used in G-code. Parameters can also be passed directly to subroutines

Description

Both global and local variables can be used in the G-code runtime application. Global variables are identified by a preceding "$" character, while local variables are declared without a preceding character or keyword. As soon as a value is assigned to a global variable, it can be accessed, read and/or changed from all parts of the program. Local variables are only effective within the program area in which a value was assigned to them for the first time.

Parameters can be transferred to subroutines via the local parameter list "(P1...P<n>)".

Example:

```
Subroutine1(P1= 10) //P1=10
Subroutine2(a, 100, „TEST“) //P1=a, P2=100, P3="TEST"
```

P1 to P<n> are reserved words for parameter transfer and cannot be used by the customer as self-defined variables.

3.1.2 Feature 914838: Extended number of M-codes per kinematic in GCO and PLC

Description

In GCO 4.6, the 100 global M-codes are replaced by 1000 kinematic-specific M-codes to synchronize signals between GCO and the PLC.

The following NC commands were added to GCO to control the local signals:

- SetLocalSignal (SLSG): Sets a local kinematic signal to "true"
- ResetLocalSignal (RLSG): Resets a local kinematic signal to "false"

- WaitForLocalSignal (WLSG): Waits until a local kinematic signal changes to "true"
- WaitForLocalSignalReset (WLSGR): Waits until a local kinematic signal changes to "false"

In addition, CXA_PLCOpen is extended by the following function blocks:

- MB_GetKinLocalSignal: Retrieves the current status of the specified local signal of the target kinematics
- MB_ResetKinLocalSignal: Resets a specified local signal of the target kinematics
- MB_KinResetLocalSignalCmd: Asynchronously resets a specified local signal of the target kinematics
- MB_SetKinLocalSignal: Sets a specified local signal of the target kinematics to "true"
- MB_KinSetLocalSignalCmd: Asynchronously sets a specified local signal of the target kinematics to "true"
- MB_KinSetLocalSignalCmdOpt: Sets a specified local signal of the target kinematics to "true" in the next command. (Sets a local signal **once**, triggered when the next Motion command is enabled)
- MB_KinWaitForLocalSignalCmd: The command waits until a specific local signal of the target kinematics is set to "true" and then resets the local signal to "false" if the input parameter "AutoReset" of function blocks is set to TRUE
- MB_KinWaitForLocalSignalResetCmd: The command waits until a specific local signal of the target kinematics is reset to "false"
- MB_HandleKinLocalSignalsMapping: Processes operations in connection with the assignment of local kinematic signals
- MB_ReadAllKinLocalSignals: Reads the status of all local signals of the target kinematics
- MB_ResetAllKinLocalSignals: Resets all local signals of the target kinematics to FALSE

3.2 Resolved defects and modifications

3.2.1 Bug 1171630: The subroutine cannot access the global variable defined in the main script

Description

The global variable defined in the main script could not be used in the subroutine.

3.3 Notes on use and known restrictions

3.3.1 Bug 1187083: A global variable defined in a subroutine cannot be accessed in the main program

Description

An error is reported due to an undefined variable if global variables defined in a subroutine are used in the main program.

Workaround

Initialize this global variable in the main program to use it there.

4 Version 4.4.0

4.1 New functions

4.1.1 Feature 1062733: G-code is extended to local variables

Description

In addition to global variables, G-code also supports local variables with dynamic data types. Local variables are defined with the keyword "LOCAL" and can be specified with/without an initial value. The scope of validity of local variables is limited to a file or a block, as shown in the example program.

Example:

```
startPos_X = 10                     // startPos_X:
                                       [10, global]
LOCAL startPos_Y = 20               // startPos_Y:
                                       [20, local in sample
                                       program]
FOR INDEX = 1 TO 3 DO
  LOCAL posX =  startPos_X + INDEX  // posX: [local
                                       inner For Statement]
  LOCAL posY =  startPos_Y + INDEX  // posY: [local
                                       inner For Statement]
  POS = posX + posY                 // POS: [global]
ENDFOR
```

4.2 Resolved defects and modifications

4.2.1 Bug 1118239: Subroutine path node does not work correctly

Description

The default path for subroutines cannot be changed via the Data Layer. When attempting to delete part of the path, the string is always incorrectly truncated at the previous slash instead of deleting only the desired characters.

Bugfixing

After saving, the default subroutine path in the Data Layer or JSON file has to display exactly the string entered by the user, without unexpected truncations or automatic trimming of the path.

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